WATER LEAK DIAGNOSIS—REMOVABLE ROOF AND CONVERTIBLE TOP—SERVICE TIPS	Article No. 02-21-14

FORD: 2002 THUNDERBIRD

ISSUE

Some vehicles may exhibit water leaks with the removable roof installed or convertible top in the raised position. This may be caused by inadequate sealing at various points in the roof area.

ACTION

Refer to the following service tips for repair information. This article will address the most common area of water leaks associated with the removable roof and convertible top and assist the technician in making an effective repair.

SERVICE INFORMATION

NOTE

THUNDERBIRD ROOF SEALING SYSTEMS ARE DESIGNED TO PREVENT WATER FROM ENTERING THE VEHICLE DURING NORMAL EVENTS OF RAIN. WATER INGRESS RESULTING FROM EXPOSURE TO A HIGH PRESSURE WASHER (POWER WASHER) OR AUTOMATED CAR WASH THAT USES HIGH PRESSURE WASHING EQUIPMENT SHOULD BE CONSIDERED NORMAL.

Removable Roof Weatherstrip Preliminary Position Check

1. Check that all weatherstrips are positioned correctly on the removable roof, vehicle header and a-pillar. If weatherstrips are positioned correctly, or leakage continues after adjustments, proceed with this article as needed by symptom. Refer to Figures 1 & 2.

NOTE

ONLY WEATHERSTRIPS THAT ARE PHYSICALLY DAMAGED SHOULD BE REPLACED. MOST WATER LEAKS CAN BE REPAIRED BY FOLLOWING THE STEPS IN THIS ARTICLE THAT PROVIDE ADJUSTMENT DETAILS.

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Removable Roof Water Leaks At A-Pillar (Front Upper Corner of Door Glass)

1. Reset the power window short drop function. Refer to the Owner's Guide for additional information.
2. Verify the water leak concern. For testing, use only normally pressurized tap water from a standard size garden hose.
3. Remove the roof and check that the weatherstrip corner at the top of the a-pillars near the windshield header weatherstrips are seated/positioned correctly. Refer to Figure 3.
   a. If gaps are present, correct by pushing/repositoning weatherstrip as needed. Reinstall the roof and water test. Stop or continue with this article as needed.
   b. If weatherstrips were seated correctly, continue with Step 4.
4. Water which enters the vehicle between the side weatherstrip and the door glass near the top of the a-pillar can be repaired by adding a seal stuffer, part number 2W6Z-76025B52-AA. The part package contains 2 stuffers. Install one stuffer on each side of the vehicle following Steps 5-9.
5. Remove the roof from the vehicle and place in the moveable cart.
6. Dry all moisture and clean all dirt and debris from the weatherstrips on the roof.
7. Apply 3-4 small evenly placed dabs of Ford Trim & Weatherstrip Adhesive, part number TA-14, to one side the seal stuffer. Do not coat the entire stuffer with adhesive.
8. Position the seal stuffer behind the secondary seal approximately 1.25 inches rearward from the front edge of the seal. For best results, have a helper open the weatherstrip so proper placement can be obtained. Refer to Figure 4. The primary weatherstrip has been removed in the figure for clarity.
9. Install the roof and water test to verify the repair.
10. If water continues to enter the vehicle in the same location, the weatherstrips may require adjustment or the door glass may not be fully rolling up into the weatherstrip. Continue with the next section, Removable Roof Leaks Around Door Glass Weatherstrips.

Removable Roof Leaks Around Door Glass Weatherstrips

1. Water which enters the vehicle between the side weatherstrips and the door glass may be repaired by adjusting or repositioning the weatherstrips. First, reset the power window short drop function. Refer to the Owner's Guide for additional information.
2. Check that the weatherstrips along the glass are uniformly pressing against the door glass. This can be done by moistening the weatherstrips with water and closing the door. The wet weatherstrip will clearly show their contact pattern against the glass.
3. Where no contact pattern is present, firmly grasp and pull the weatherstrip to the desired location as to achieve an adequate, uniform seal against the glass. The weatherstrip is attached to metal flanges on the roof which will bend slightly.
4. If adjustments do not correct the condition, the door glass may not be fully rolling up into the weatherstrip. Continue with Step 5.
5. Make certain the windows are capable of completely seating into the weatherstrip.
   a. Open the door and manually trip the door latch closed. This will raise the glass with the short drop function.
   b. Measure the distance between the peaked edge on the rear of the door glass to the top of the belt molding seal. If greater than 3/8 inch, the glass is not rolling up fully into the weatherstrips, proceed to step 6. If 3/8 inch or less, the glass is lifting properly. Check again for mispositioned or damaged weatherstrips. Refer to Figure 11.
6. The glass may not mechanically roll up far enough to seal.
   a. Keep door open, and the window in the raised position from the short drop feature.

b. Remove the door trim panel to access the rear glass adjustment nut. Refer to Workshop Manual, Section 501-05.

c. Loosen the rear 10 mm regulator clamp bolt and pull upward on the glass until a maximum of 3/8 inch can be measured between the peaked edge on the rear edge of the door glass and the top of the belt molding seal. Refer to Figure 11.

d. Torque the bolt to 12 Nm (9 lb-ft).

e. Assemble the door trim panel, flip the latch closed to drop the glass, and water test. If water leaks are repaired, return vehicle to customer. If water leak continues, check again for mispositioned or damaged weatherstrips.

Removable Roof Water Leaks Along Windshield Header

1. Verify the two (2) roof T40 a-pillar attaching screws are torqued to 80 Lb-in (9 N•m), and that the latch handles below the porthole windows are fully engaged.
2. If fasteners were loose or latches were not engaged, water test. If still leaking, continue with article. If leak is repaired, return vehicle to customer.
3. Most header water leak concerns can be verified by directing normally pressurized tap water from a standard size garden hose over the removable roof near the header. Remove the interior header trim panel from the removable roof by grasping and firmly pulling down the panel. This may aid in locating the source area of the water leak.
4. If water is entering the vehicle during this test, it may be penetrating between the removable roof and roof header trim bead, near the edges of the roof.
5. Remove the roof and carefully set on the moveable cart.
6. Check that the weatherstrip corners at the top of the a-pillars near the windshield header weatherstrips are seated/positioned correctly. Refer to Figure 3.

a. If gaps are present, correct by pushing/repositioning the weatherstrip as needed. Reinstall roof and water test. Stop or continue with this article as needed.

b. If weatherstrips were seated correctly, continue with Step 7.

7. On the side of the roof where the leak is present, apply a small bead of clear silicone based caulk starting from the edge of the roof and move 8 inches inboard towards the center. Refer to Figure 5.

8. Use a soft, flexible plastic or rubber paddle to force the silicone between the removable roof and the roof header trim bead, filling any gaps.

CAUTION
DO NOT USE TOOLS TO DIRECT THE SILICONE THAT WILL DAMAGE THE PAINT. TEST TOOL ON A SCRAP PIECE OF PAINTED SHEET METAL FIRST TO CHECK FOR EVIDENCE OF PAINT DAMAGE.

9. Immediately clean the excess silicone from the roof with a damp cloth.

10. Repeat the application of silicone on the inside seam between the removable roof and roof header trim bead. Refer to Figure 6.

11. Install the roof and reset the power window short drop function. Refer to the Owner's Guide for additional information.

12. Water test and verify the repair.

Convertible Top Water Leaks Around Door Glass

1. Reset the power window short drop function. Refer to the Owner's Guide for additional information.

2. Verify the water leak concern. Direct normally pressurized tap water from a standard size garden hose around the effected door glass.

3. Check that the weatherstrips up the a-pillar, at the corner of the a-pillar and windshield and along the convertible top that seals the door glass are not damaged and are positioned correctly. Adjust/reposition as needed and water test. Stop or continue with this article as needed. Refer to Figures 7-10.

NOTE
ONLY WEATHERSTRIPS THAT ARE PHYSICALLY DAMAGED SHOULD BE REPLACED. MOST WATER LEAKS CAN BE REPAIRED BY FOLLOWING THE STEPS IN THIS ARTICLE THAT PROVIDE ADJUSTMENT DETAILS.

4. Roll down the windows and operate the convertible top down and back up. Once the top is at the windshield header and the pump motor begins to "stall", keep the button depressed for 2-3 seconds additional before latching the top. If the top is not completely positioned up, the top frame and weatherstrips will not be not positioned correctly. Roll the windows up and water test. Stop or continue with this article as needed.

5. Check for excessive gaps between the weatherstrips on the convertible top. Adjust the position of the weatherstrips on their mounting flange, or bend the mounting flange as needed to correct. Stop or continue with this article as needed. Refer to Figures 7-10.

6. Lower the convertible top and check that the weatherstrip corner at the top of the a-pillars near the windshield header weatherstrips are seated/positioned correctly. Refer to Figure 3.

a. If gaps are present, correct by pushing/repositioning the weatherstrip as needed. Raise the roof and water test. Stop or continue with this article as needed.

b. If weatherstrips were seated correctly, continue with Step 7.

NOTE
IN MOST CASES, IF THE WINDOW IS ALLOWED TO ROLL COMPLETELY UP INTO THE WEATHERSTRIP, ALL CONTACT POINTS ALONG THE GLASS WILL BE ACCEPTABLE.

7. Make certain the windows are capable of completely seating into the weatherstrip.

a. Raise the roof. Open the door and manually trip the door latch closed. This will raise the glass with the short drop finction.

b. Measure the distance between the peaked edge on the rear of the door glass to the top of the belt molding seal. If greater than 3/8 inch, the glass is not rolling up fully into the weatherstrips, proceed to Step 11. If 3/8 inch or less, proceed to Step 8. Refer to Figure 11.

8. Lift the outside door handle and trip the door latch open which will drop the door glass. Close the effected door and measure and record the distance between the silkscreen on the door glass and the top of the belt molding seal. The measurement distance is not critical. The rule in the figure is for reference. Refer to Figure 12.
9. Open the effected door. Exert light hand pressure against outside of the glass. Shut the door with normal force and keep pressure on the glass until the short-drop function raises the glass into the weatherstrip. Again measure between the silkscreen and top of the belt molding seal. Refer to Figure 12.
10. If measurement is now greater than observed in Step 8, go to Step 12. If measurement remained the same, go to Step 11.
11. The glass may not mechanically be able to roll up far enough to seal.
    a. Raise the roof. Open the effected door and command the system to roll the glass up with the short-drop feature. This is done by manually tripping the door latch.
    b. Remove the door trim panel to access the rear glass adjustment nut. Refer to Workshop Manual, Section 501-05.
    c. Loosen the rear 10mm regulator clamp bolt and pull upward on the glass until a maximum of 3/8 inch can be measured between between the peaked edge on the rear of the door glass to the top of the belt molding seal.
    d. Torque the bolt to 9 Lb-ft (12 N•m).
    e. Assemble the door trim panel, flip the door latch open to drop the glass and water test. If water leaks are repaired, return vehicle to customer. If water leak continues, proceed to Step 12.

12. The window motor may be stalling before the glass can properly seal. Strike "down and in" along the stitch bead on the edge of the convertible top, starting from the a-pillar area back approximately 20 inches with a rubber mallet. This area is where the top of the glass penetrates a U-shaped channel in the weatherstrip. If the weatherstrip flange on the convertible top is not rolled inboard enough, the door glass will not properly enter the weatherstrip. Refer to Figure 13.
13. After this adjustment is made, normally open and close the door checking that the glass is rolling all the way up into the weatherstrip. Water test. Stop or continue this article at Step 14.
14. If the glass still is not fully entering the weatherstrips, adjust the channel in the rear portion of the second weatherstrip "up" to allow closure. This area of the weatherstrip is where it transitions from the channel along the roof to a compression seal along where the B-pillar would be present. Use the handle of a rubber mallot to bend the weatherstrip and convertible top flange "up". Water test. Stop or continue this article as needed. Refer to Figure 14.
15. Check that the weatherstrips are uniformly pressing against the door glass. This can be done by moistening the weatherstrips with water and closing the door. The wet weatherstrips will clearly show their contact pattern against the glass. This is critical along the rearward slope of the glass (B-pillar).
16. The weatherstrips are positioned on metal mounting flanges that are part of the convertible top frame. These flanges may be bent slightly to improve the weatherstrip pattern. To adjust, simply grasp the weatherstrip and move it inboard or outboard to improve the contact against the glass.

PART NUMBER	PART NAME
TA 14	Weatherstrip Adhesive
2W6Z-76025B52-AA	Hard Top Seal Stuffer
F7AZ-19554-CA	Clear Silicone Rubber

OTHER APPLICABLE ARTICLES: NONE
WARRANTY STATUS: INFORMATION ONLY
OASIS CODES: 102000, 105000, 110000, 701000

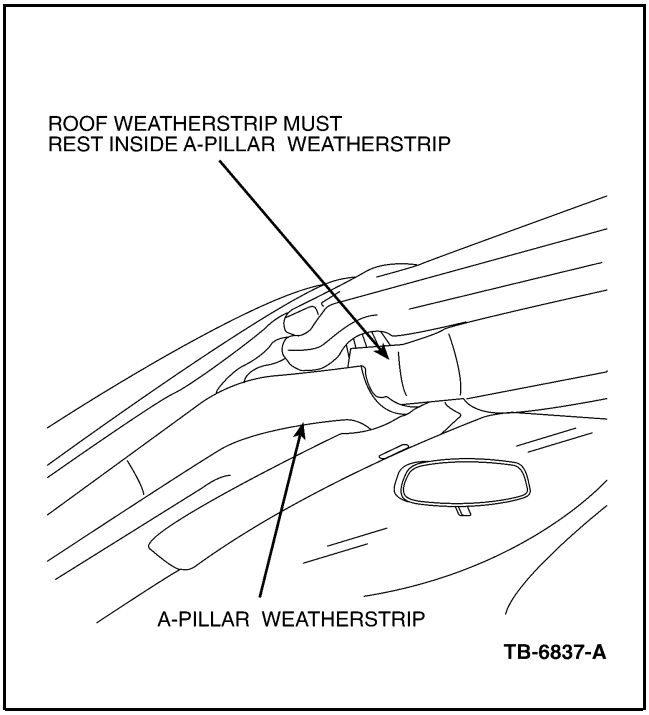


Figure 1 - Article 02-21-14

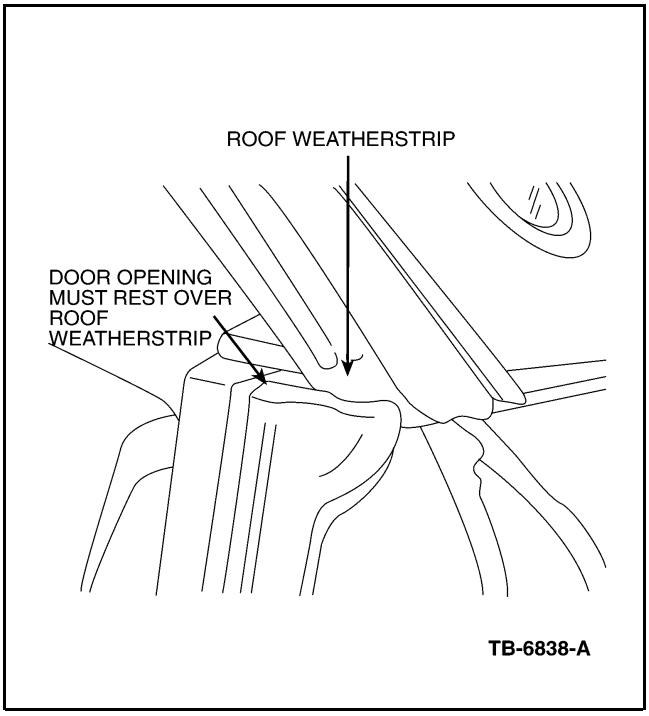


Figure 2 - Article 02-21-14

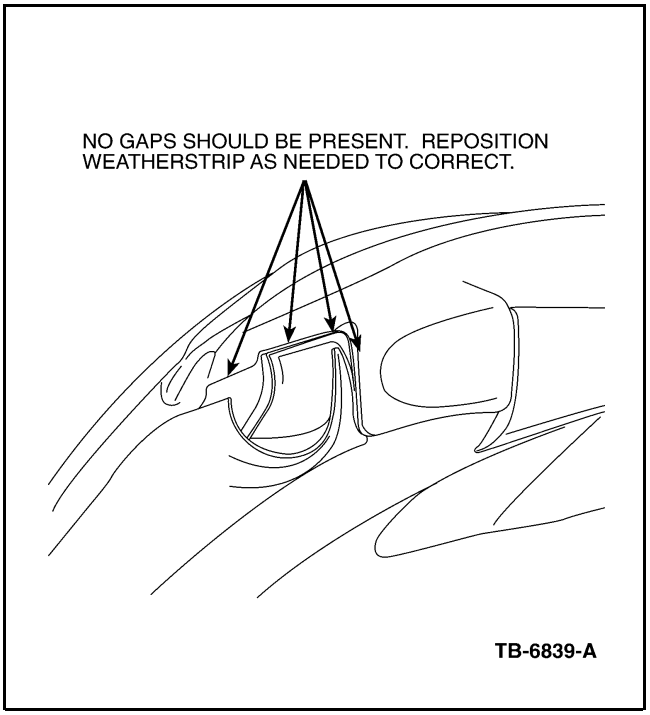


Figure 3 - Article 02-21-14

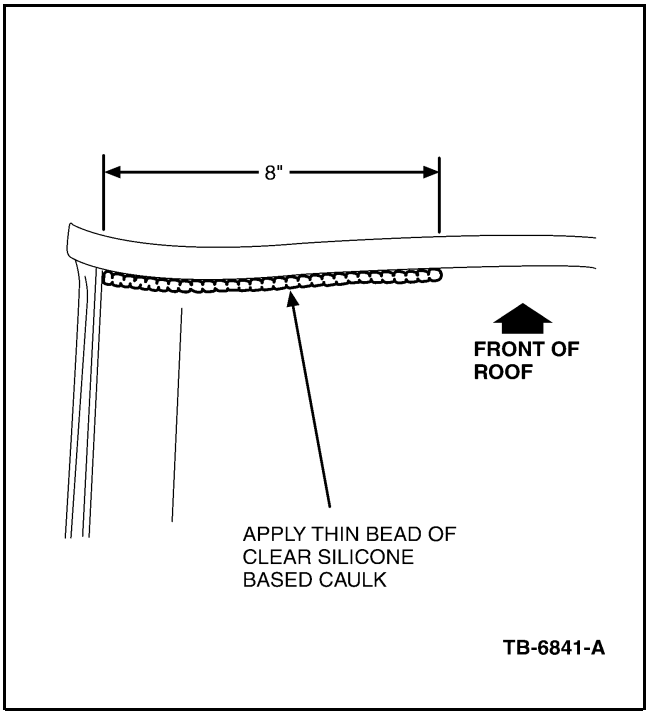


Figure 5 - Article 02-21-14

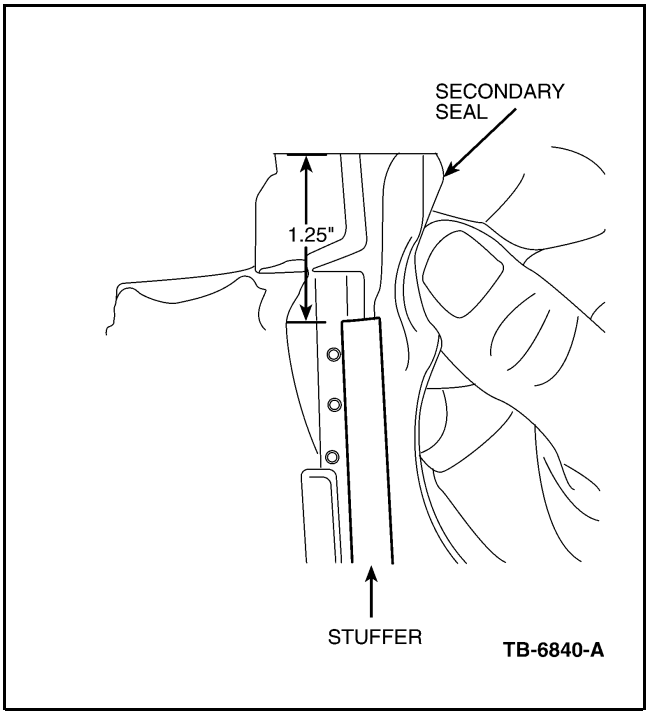


Figure 4 - Article 02-21-14

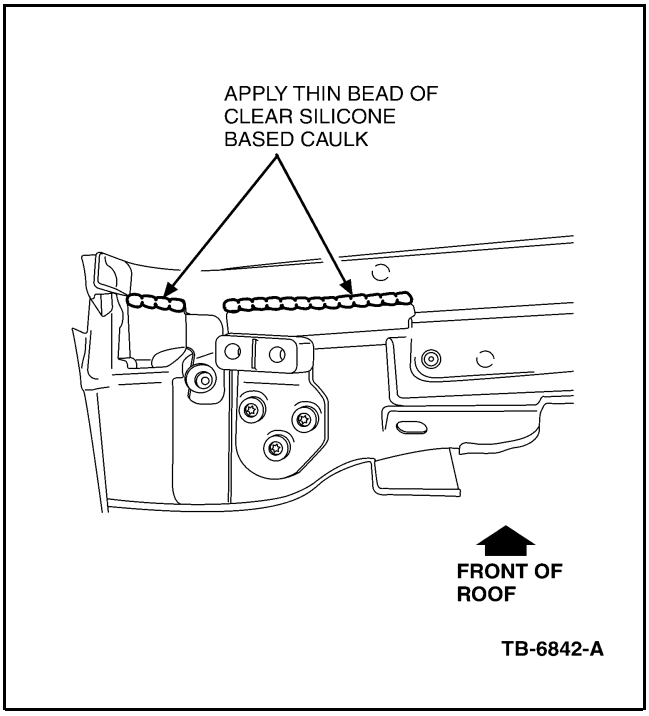


Figure 6 - Article 02-21-14

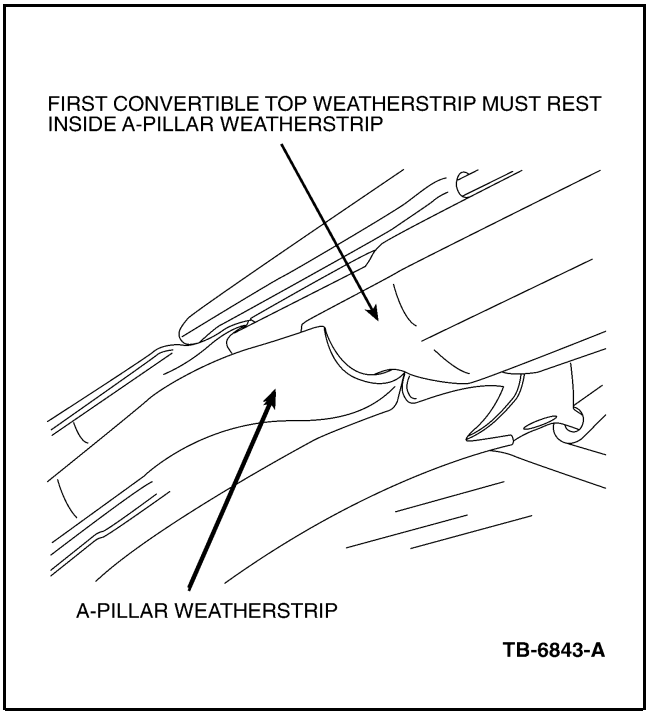


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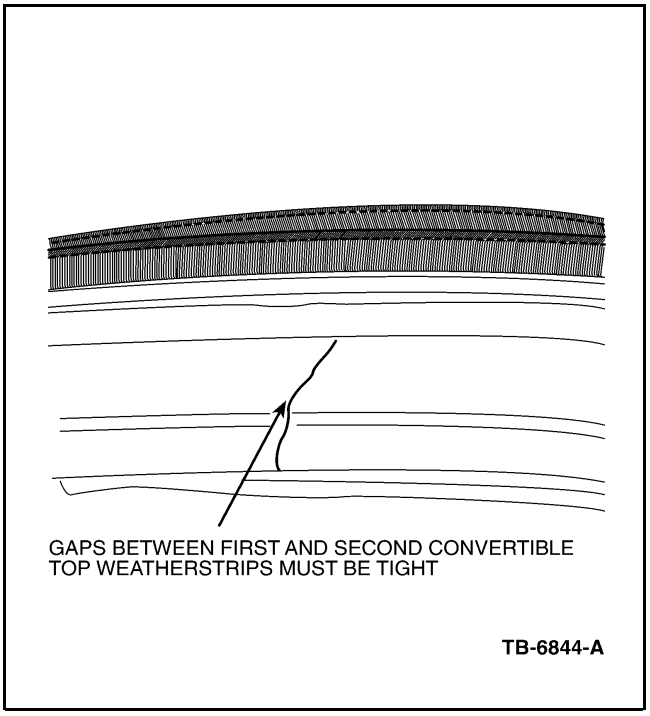


Figure 8 - Article 02-21-14

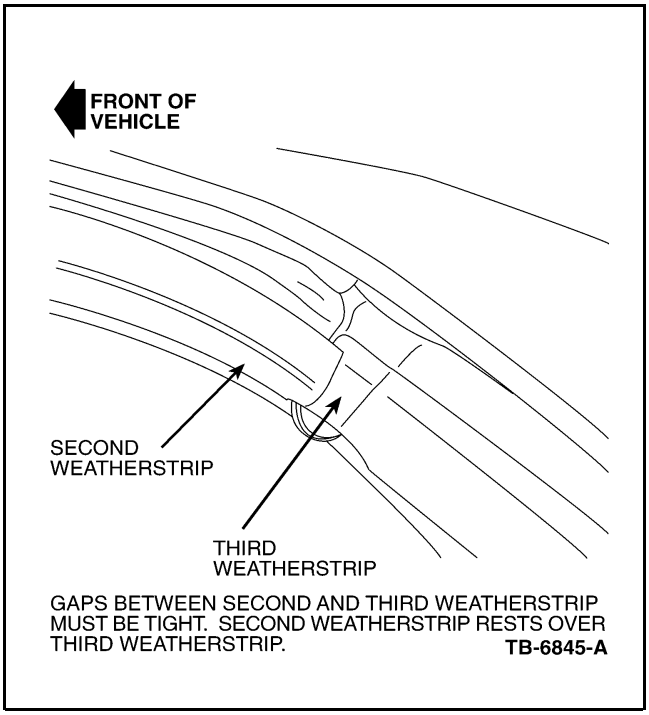


Figure 9 - Article 02-21-14

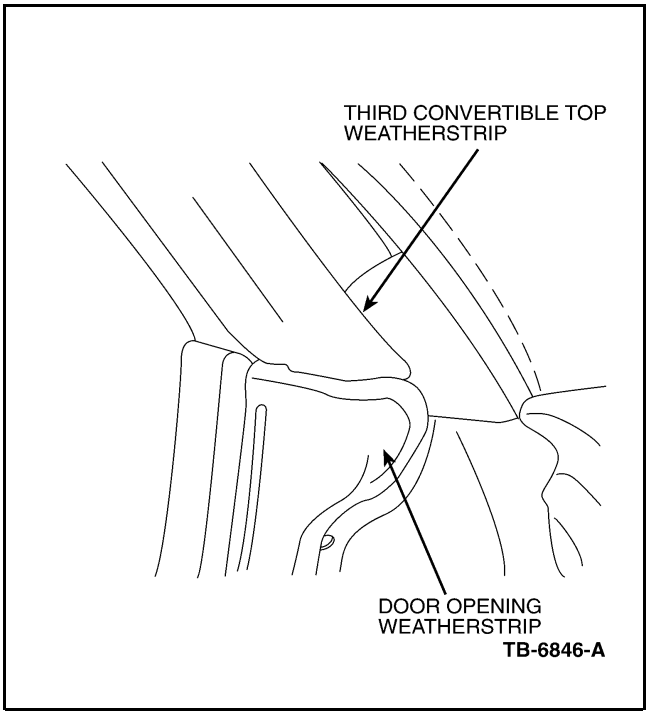


Figure 10 - Article 02-21-14

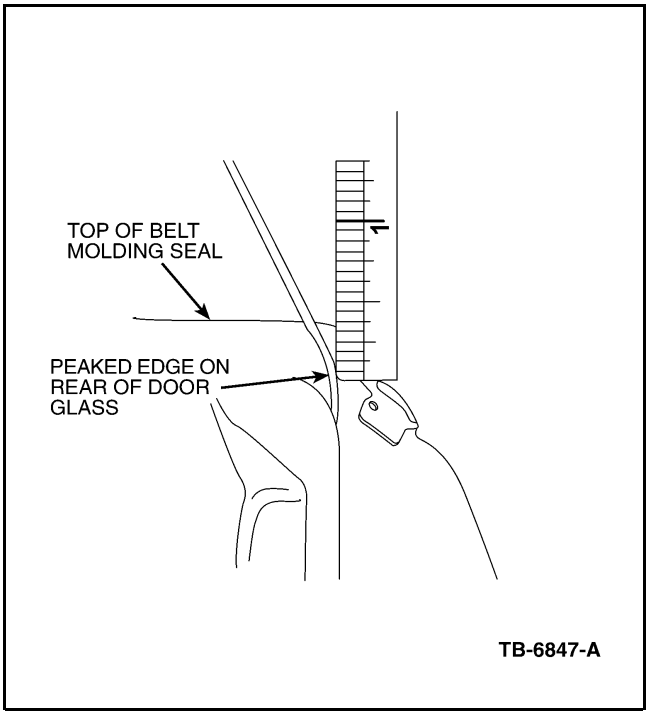


Figure 11 - Article 02-21-14

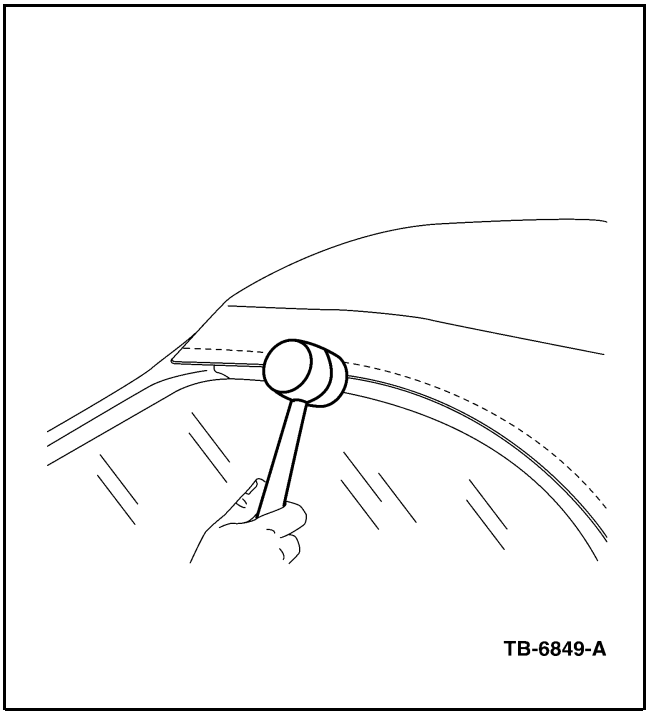


Figure 13 - Article 02-21-14

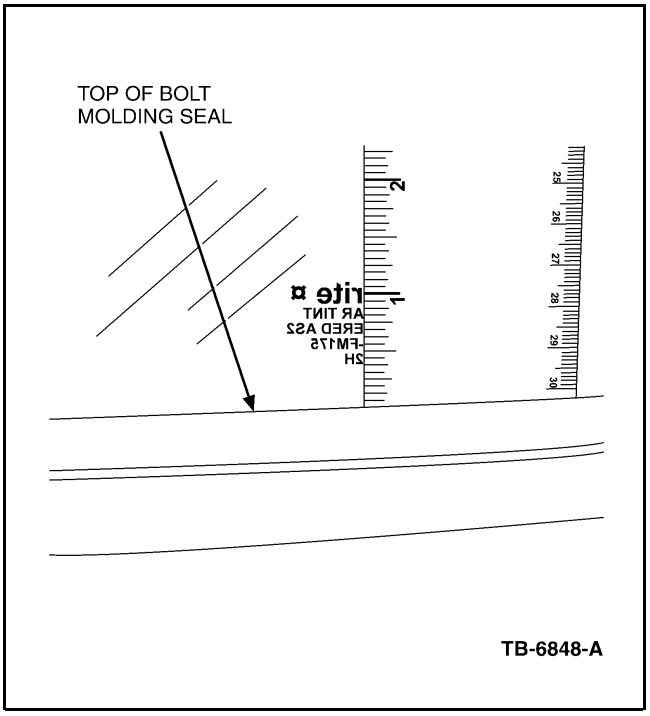


Figure 12 - Article 02-21-14

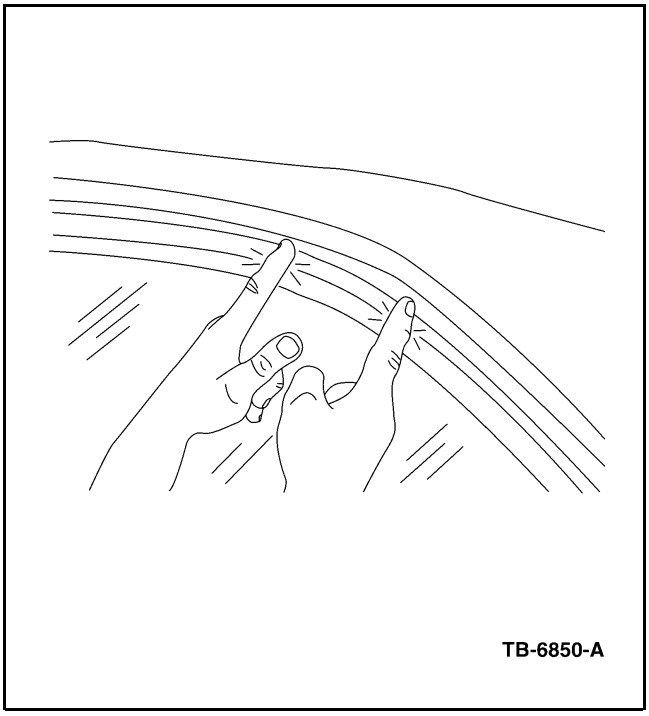


Figure 14 - Article 02-21-14

NOTE: The information in Technical Service Bulletins is intended for use by trained, professional technicians with the knowledge, tools, and equipment to do the job properly and safely. It informs these technicians of conditions that may occur on some vehicles, or provides information that could assist in proper vehicle service. The procedures should not be performed by "do-it-yourselfers". Do not assume that a condition described affects your car or truck. Contact a Ford, Lincoln, or Mercury dealership to determine whether the Bulletin applies to your vehicle.